Hotel Bookings Analysis Report

Observed Patterns and Trends :--

1. Meaningful Trends and Patterns in the Data

- Around 20% of total bookings are cancelled, showing cancellations are a consistent part of the dataset.
- A strong positive relationship exists between selling price, cost price, and booking value, indicating a stable pricing structure.
- Booking value varies across months, showing clear seasonal fluctuations with certain peak and low periods.
- Cancellation rates are not uniform across segments, with higher values observed in Travel Agent bookings and Standard room types.

2. How do booking pattern vary across channels, room types, and star ratings

- **Across Channels:**
  Travel Agent bookings show the highest cancellation rates, Mobile App is moderate, and Web bookings have the lowest cancellations.
- **Across Room Types:**
  Standard rooms have the highest cancellations, Suites are moderate, and Deluxe rooms show the lowest cancellation levels.
- **Across Star Ratings:**
  4–5 star hotels show slightly higher cancellations compared to 2–3 star hotels, though the difference is not very large.

3. Some things noticed during cancellation behavior

- Around 20% of bookings are cancelled, indicating a noticeable cancellation level.
- Cancellations are higher in Travel Agent bookings and lower in Web bookings.
- Standard room types show higher cancellations compared to Deluxe rooms.
- Slightly higher cancellations are observed in 4–5 star hotels.
- Overall, cancellations are concentrated in specific segments rather than evenly distributed.

Root Cause Explanations :--

1. What might be the reasons for the observed cancellation patterns

- Travel Agent bookings: Higher cancellations likely due to bulk or tentative bookings with greater flexibility.
- Standard room types: More cancellations as these attract price-sensitive customers who may switch options.
- Higher star hotels (4–5): Higher cancellations possibly due to higher prices and bookings made well in advance.
- Discount-driven bookings: Presence of coupons/cashback may lead to less committed bookings.
- Seasonal effects: Changes in demand across months may influence booking confirmations and cancellations.

2. Why do some booking channels or property types perform better than others

- Web channel performs better due to more direct and intentional bookings, leading to lower cancellations.
- Travel Agent channel shows weaker performance due to bulk/tentative bookings with higher cancellation likelihood.
- Deluxe rooms perform better as they attract more committed customers, resulting in lower cancellations.
- Standard rooms show weaker performance due to price-sensitive behavior and higher booking volatility.

3. Seasonal or temporal trends influencing booking values or stay lengths

- Booking values vary across months, showing clear seasonal fluctuations.
- Certain months have higher average booking values, while others show noticeable dips.
- This indicates changes in travel demand over time.
- Where available, stay lengths also show variation, suggesting differences in travel patterns across periods.

Actionable Business Reccomendations :--

1. Strategies to reduce cancellations

- Implement stricter cancellation policies (e.g., partial refund or time-based charges).
- Promote direct (Web/App) bookings, which show lower cancellation rates.
- Offer incentives for confirmed stays (e.g., cashback after completion).
- Send booking reminders and confirmations closer to travel dates.
- Limit high-risk bookings from channels with higher cancellation rates.

2. Recommendation to Improve Profitability & Repeat Bookings

- Increase share of direct (Web/App) bookings to reduce dependency on high-cancellation channels.
- Use targeted offers for repeat customers (loyalty, cashback on completed stays).
- Focus on upselling (e.g., upgrade from Standard to Deluxe rooms).
- Optimize discounts by limiting excessive coupon/cashback usage.

3. Identified opportunities to optimize pricing, promotions, or channel strategy

- Apply dynamic pricing based on demand (higher in peak months, lower in off-season).
- Use selective promotions instead of blanket discounts.
- Allocate inventory strategically across channels, prioritizing high-performing channels.
- Adjust pricing and offers based on room type demand and booking patterns